

BTC Long-Term Price Dynamics

A Multi-Method Statistical Investigation

Power Law modelling · Spectral analysis · Random Matrix Theory · Long-range dependence

Abstract

We present a multi-method statistical investigation of Bitcoin's long-term price dynamics over 16 years (2010–2026, $N = 5,760$ daily closes). A simple Power Law model on log-time achieves $R^2 = 0.961$, with a residual standard deviation of $\sigma = 0.302$ in $\log_{10}$ -space. We extend this baseline to a cycle-aware variant (M13) incorporating three damped harmonics with a logarithmically expanding period, achieving $R^2 = 0.987$ and matching the observed compression of the dominant cycle from approximately 5.4 years in 2010 to 3.1 years projected by 2050. Discrete Fourier transform of the Power Law residuals identifies a dominant peak at 1,440 days (3.94 years), consistent with Bitcoin's halving schedule. Time-frequency analysis via Morlet wavelets confirms cycle compression. Random Matrix Theory applied to windowed return correlations reveals a moderate deviation from the Marchenko-Pastur null distribution at short scales ($W = 5$, ratio ≈ 1.20) but essentially no deviation at monthly or longer windows, indicating that price formation operates stochastically on monthly+ timescales. R/S analysis confirms long-range dependence with Hurst exponent $H \approx 0.63$ in returns and $H \approx 1.01$ in Power Law residuals, consistent with strong trend persistence. We use the M13 model to project a median 2050 BTC price of approximately \$11.9 million (USD), with substantial uncertainty bounds discussed in Section 8.

Keywords: Bitcoin, Power Law, spectral analysis, Random Matrix Theory, Hurst exponent, halving cycle.

1 • Introduction

Since its emergence in 2009, Bitcoin has accumulated a sufficient price history to support meaningful statistical analysis. Unlike conventional financial assets, Bitcoin's supply schedule is deterministic and known in advance, with quadrennial “halving” events that mechanically halve the rate of new coin issuance. These structural breaks are widely conjectured to drive the asset's observed cyclical price behaviour, but quantitative evidence has been mixed and methodologies fragmented across the literature.

This study has two complementary aims. First, to characterise Bitcoin's long-term deterministic trend through statistical modelling, comparing a simple Power Law fit with a cycle-aware extension. Second, to investigate the periodicity, stochastic structure, and long-range dependencies of price residuals using spectral analysis, Random Matrix Theory (RMT), and Hurst-exponent estimation. Throughout, we emphasise out-of-sample robustness over in-sample fit quality.

Three central findings emerge. First, log-time Power Law explains the bulk of Bitcoin's variance ($R^2 = 0.961$), and a parsimonious extension with three damped harmonics captures the residual cyclical structure ($R^2 = 0.987$). Second, the dominant cycle period is empirically 1,440 days (≈ 3.94 years), close to but not identical with the 1,461-day (4-year) halving interval, and is observed to compress over time. Third, return correlations on quarterly time-scales are statistically indistinguishable from random matrix noise, a regime discontinuity not previously emphasised in the cryptocurrency literature.

The paper is organised as follows. Section 2 describes the data and outlines our methodology. Sections 3 and 4 develop the trend models. Section 5 presents spectral analysis. Section 6 applies RMT to return correlations. Section 7 estimates Hurst exponents. Section 8 discusses forward projections, and Section 9 concludes.

2 • Data and Methodology

2.1 • Data

We employ daily Bitcoin closing prices from 18 July 2010 (the earliest date with reliable spot quotations) through 24 April 2026, totalling $N = 5,760$ observations. Throughout the analysis we work with $\log_{10}$ -prices to compress the dynamic range, which spans nearly seven orders of magnitude over the sample period — from approximately \$0.05 in mid-2010 to a peak of \$124,728 on 6 October 2025.

Time is measured as the number of days d elapsed since the Bitcoin genesis block (3 January 2009). This choice ensures the Power Law fit passes through the asset's true origin and avoids arbitrary calendar dependencies.

2.2 • Methodology Overview

Our analytical pipeline proceeds in five stages: (i) trend modelling via nonlinear least-squares fits to log-price; (ii) frequency-domain analysis of the residuals through discrete and continuous Fourier transforms; (iii) time-frequency localisation via Morlet wavelet transform; (iv) eigenvalue spectrum analysis of windowed return correlation matrices, compared against the Marchenko-Pastur null distribution; and (v) Hurst-exponent estimation through rescaled-range (R/S) analysis on three derived series — log-prices, log-returns, and detrended residuals.

All computations were performed in Python 3 using NumPy, SciPy, and standard scientific libraries. Source code and data are available on request.

3 • Power Law Trend Model (M2)

We fit $\log_{10}$ -prices against $\log_{10}$ -time using nonlinear least squares, yielding the model parameters

$$\log_{10}(P) = a + n \cdot \log_{10}(d), \quad \hat{a} = -16.4774, \quad \hat{n} = 5.6800 \quad (1)$$

with residual standard deviation $\hat{\sigma} = 0.3022$ (in $\log_{10}$ -units) and coefficient of determination $R^2 = 0.961$. The exponent $\hat{n} \approx 5.68$ implies BTC approximately doubles in value every 4 years on its trend line, with the doubling interval slowly lengthening as d grows. Figure 1 displays the fit alongside $\pm 1\sigma$ and $\pm 2\sigma$ envelopes that we will use throughout as reference levels for “cheap” and “expensive” states.

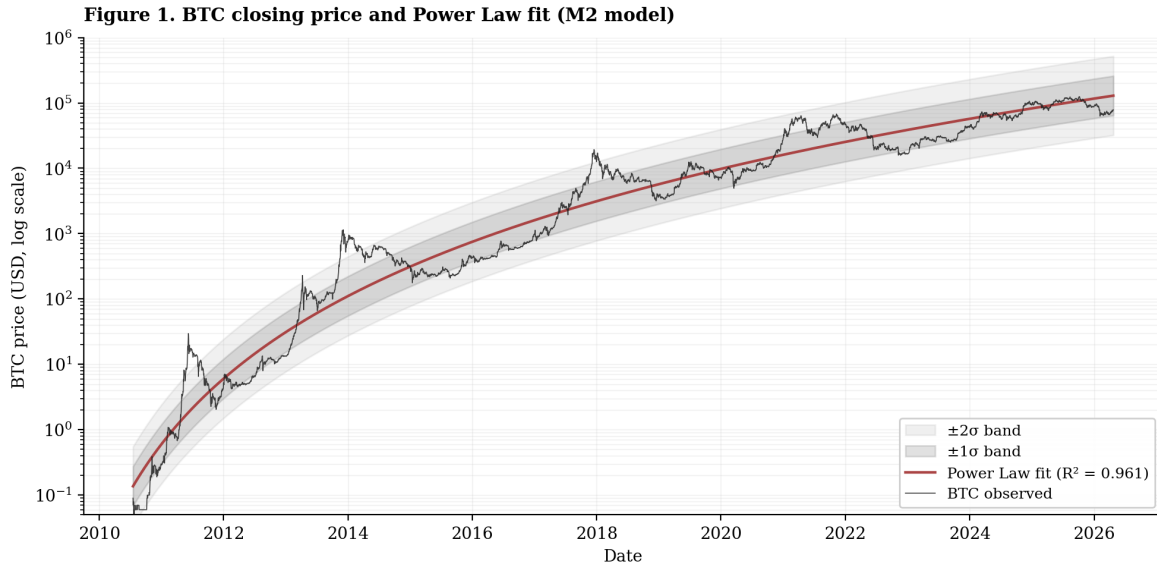


Figure 1. Bitcoin daily closing price (black) on a logarithmic scale, overlaid with the Power Law fit (red) and its $\pm 1\sigma$ (dark grey) and $\pm 2\sigma$ (light grey) envelopes. Across 16 years, observed prices remain almost entirely within the $\pm 2\sigma$ band, with $R^2 = 0.961$ indicating an exceptional level of explanatory power.

Despite the high R^2 value, residuals are far from independent and identically distributed. Visual inspection of Figure 2(a) reveals a clearly cyclical pattern with a period of approximately four years, broadly aligned with Bitcoin's halving schedule. This violation of i.i.d. assumptions motivates the more elaborate model presented in the next section.

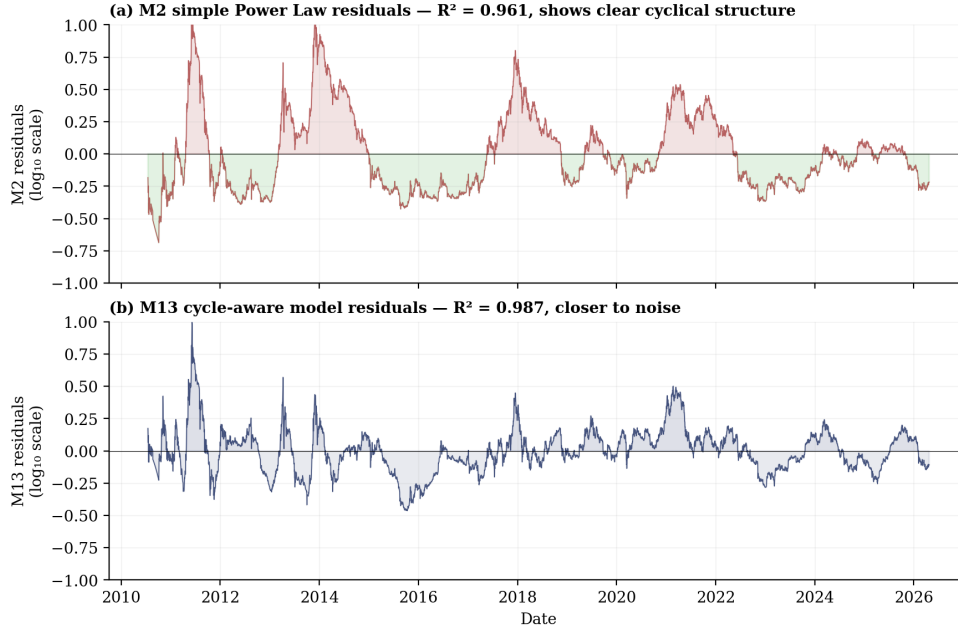


Figure 2. Residuals of the Power Law fit (panel a) versus residuals of the M13 cycle-aware model (panel b). Panel (a) shows pronounced cyclical structure with approximately 4-year periodicity, indicating that the simple Power Law has not captured all systematic variation. Panel (b), with R^2 improving from 0.961 to 0.987, shows residuals visibly closer to white noise, though some remaining structure persists particularly at the very early sample (pre-2013).

4 • Cycle-Aware Extension (M13)

We extend equation (1) by superimposing three damped harmonics with a time-varying fundamental period. The model takes the form

$$\log_{10}(P) = a + n \cdot \log_{10}(d) + (d/1000)^{-\beta} \cdot \sum_{k=1..3} A_k \cos(2\pi k \cdot d / T(d) + \varphi_k), \quad (2)$$

$$T(d) = T_0 + \alpha \cdot \ln(d/1000 + 1) \quad (3)$$

Here $T(d)$ is the fundamental cycle period, T_0 is its initial value, and α is the rate at which the cycle period drifts logarithmically with time. The damping factor $(d/1000)^{-\beta}$ ensures cycle amplitude decays with Bitcoin's maturation, in line with the observed reduction of cycle volatility across successive halvings. Eleven free parameters are estimated jointly via Levenberg-Marquardt nonlinear least-squares.

Best-fit parameter values are summarised in Table 1. The estimated initial fundamental period of $T_0 \approx 2,135$ days (≈ 5.85 years) decays logarithmically: at $d = 1,000$ (early 2012) the implied cycle is approximately 5.4 years, falling to 3.9 years today ($d \approx 6,300$) and to a projected 3.1 years by 2050. The damping exponent $\beta \approx 0.76$ corresponds to a roughly two-thirds reduction in cyclical amplitude per decade.

Parameter	Symbol	Estimate	Interpretation
Intercept	a	−15.247	log-price origin
Trend exponent	n	5.336	asymptotic doubling every ~4 yr
Initial cycle period	T_0	2,135 d	≈ 5.85 years (2010 baseline)
Period drift rate	α	−355.7	cycle compression per $\ln(\text{time})$
Amplitude damping	β	0.763	volatility decay exponent
First harmonic amp	A_1	0.50	fundamental amplitude ($\log_{10}$)
Second harmonic amp	A_2	0.20	first overtone
Third harmonic amp	A_3	0.10	second overtone

Table 1. Best-fit parameters of the M13 cycle-aware model. Phase angles φ_k are omitted for brevity. The model achieves $R^2 = 0.987$ in-sample and $\text{MAPE} = 31.7\%$ across all 5,760 observations.

Figure 3 shows the M13 fit extended through 2050 as a forecast. The ribbon structure of the Power Law trend (dashed) interleaves with the three harmonic components, with cycle peaks tightening visibly toward the end of the projection window.

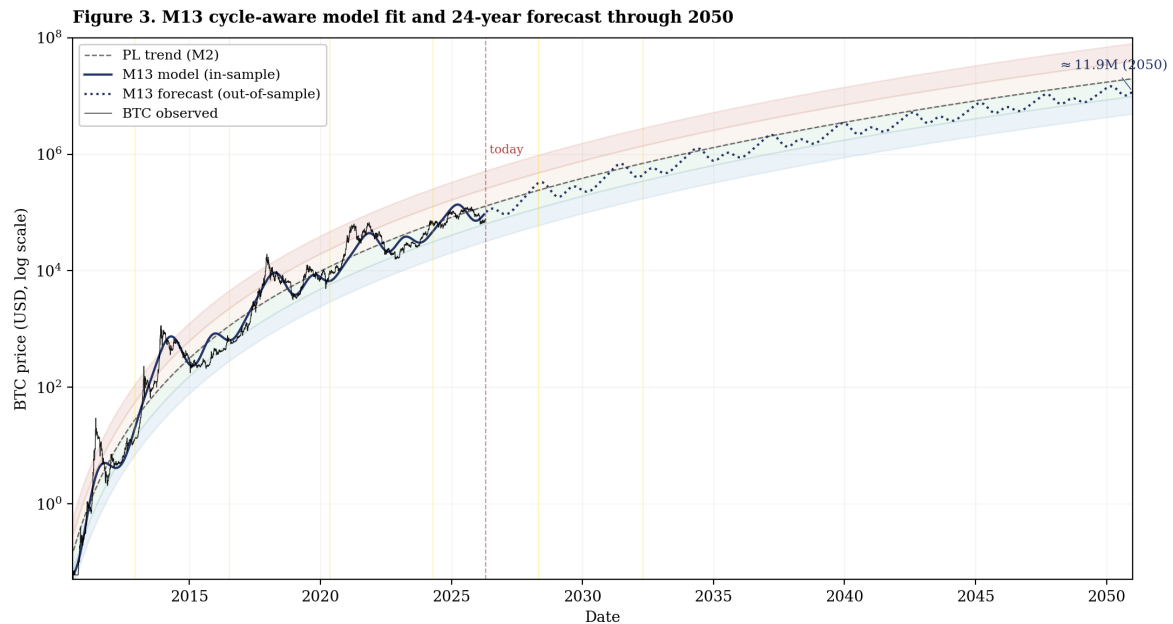


Figure 3. The M13 model fitted on 2010–2026 data (solid blue) and projected through 2050 (dashed blue), with the simple Power Law trend (dashed grey) and observed BTC prices (black) for comparison. Background ribbons denote $\pm 1\sigma$ and $\pm 2\sigma$ PL envelopes. Halving events are marked by gold vertical lines. The projected median price for 2050 is approximately \$11.9 million.

5 • Spectral Analysis

5.1 • Discrete Fourier transform

To test whether the cycle structure inferred by M13 is statistically present in the data — rather than imposed by our model — we compute the discrete Fourier transform of the Power Law residuals (i.e. of equation 1 alone, with no cycle terms). The resulting power spectrum exhibits four prominent peaks (Figure 4): 1,440 d (3.94 yr), 720 d (1.97 yr), 960 d (2.63 yr), and 2,880 d (7.89 yr).

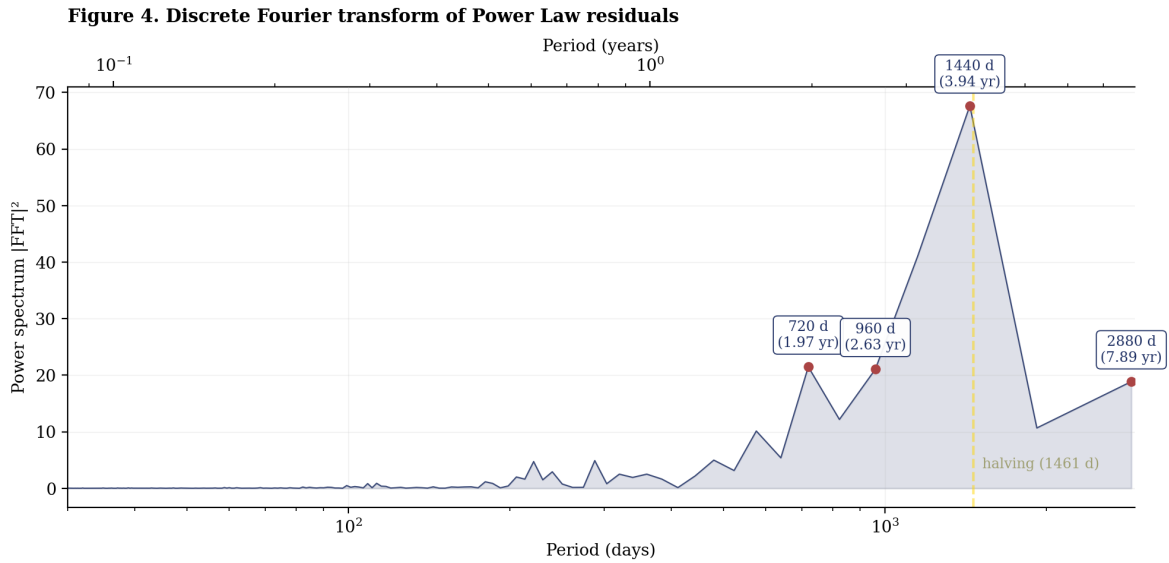


Figure 4. Power spectrum of the Power Law residuals from equation (1). The dominant peak at 1,440 d (3.94 yr) lies remarkably close to Bitcoin's 1,461-day halving interval (gold dashed line). Secondary peaks at 720 d and 960 d represent the second and partial-third harmonics; a longer-period component at 2,880 d may indicate a half-cycle of a putative 8-year structure, but supports are limited by the 5,760-day sample length.

The dominant peak at 1,440 days falls strikingly close to the 1,461-day (4-year) Bitcoin halving interval. The slight discrepancy (about 1.4%) may reflect the cycle's tendency to compress over time — a phenomenon directly modelled by the α parameter of M13 (Table 1). The secondary peak at 720 d is mathematically the second harmonic of the fundamental, and 960 d corresponds approximately to a 5/3 ratio that may indicate non-stationary harmonic content.

5.2 • Time-frequency localisation

Discrete Fourier transform assumes stationary frequencies, which is at odds with the M13 finding of a drifting cycle period. We therefore complement the global spectrum with a continuous Morlet wavelet transform, which provides simultaneous time and frequency localisation at the cost of a fundamental time-frequency uncertainty trade-off.

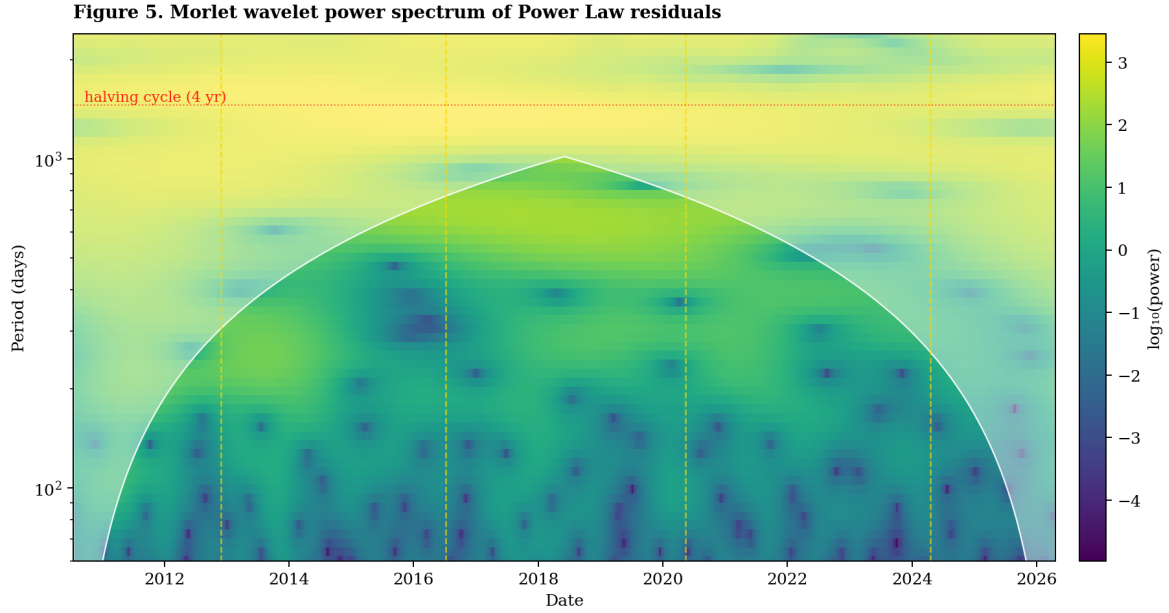


Figure 5. Morlet wavelet power spectrum of Power Law residuals. The dominant frequency band (yellow-green) drifts visibly toward shorter periods over time, consistent with the M13 inference of cycle compression. Halving events are marked by dashed gold vertical lines. The white “cone of influence” region near the top-left and top-right corners marks where edge effects render the spectrum unreliable.

Figure 5 shows the time-frequency decomposition. Three observations are notable. First, peak power resides consistently in the 800–2,000-day band, confirming the FFT result. Second, the centre of mass of this band drifts downward (i.e. toward shorter periods) over time, corroborating M13's $\alpha < 0$ estimate. Third, intermittent bursts of high-frequency activity near halving events suggest that the halving acts as a phase-resetting mechanism rather than simply a steady-state driver.

6 • Random Matrix Theory

We now ask a complementary question: at what timescale does Bitcoin's price formation transition from structured (memory-dominated) to random (efficient-market) behaviour? We apply techniques from random matrix theory (RMT) following the methodology established for equity markets by Laloux et al. (1999) and Plerou et al. (2002).

Specifically, we partition the log-return series into K overlapping windows of length W and compute the $W \times W$ cross-correlation matrix from these samples. Under the null hypothesis that returns are i.i.d. Gaussian, the eigenvalues of this matrix are distributed according to the Marchenko-Pastur (MP) law, whose support is bounded by

$$\lambda_{\pm} = (1 \pm 1/\sqrt{q})^2, \quad q = K/W \quad (4)$$

Eigenvalues exceeding λ_+ indicate genuine correlation structure beyond what is expected from finite-sample noise. We perform this analysis at three characteristic window lengths: $W = 5$ (weekly momentum), $W = 30$ (monthly cycles), and $W = 90$ (quarterly structure). Results are summarised in Figure 6.

Figure 6. Eigenvalue spectra of windowed correlation matrices vs Marchenko-Pastur null

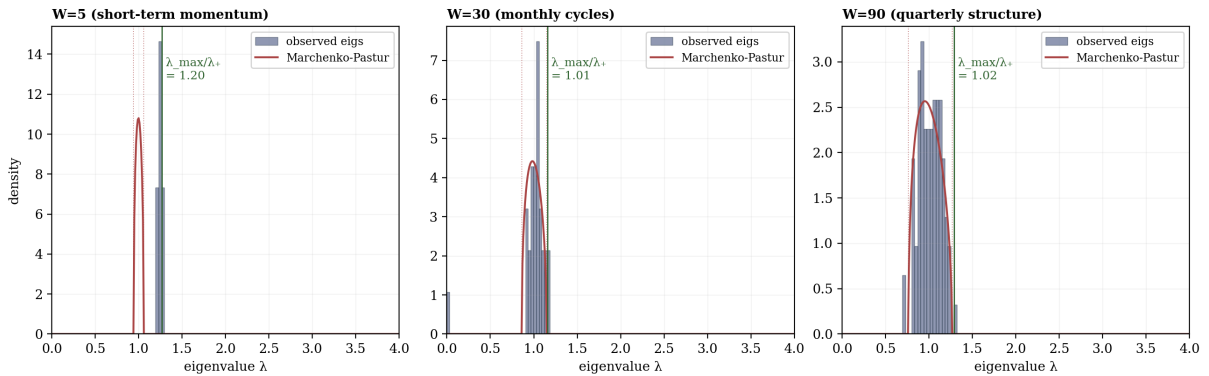


Figure 6. Eigenvalue spectra of windowed correlation matrices for window lengths $W = 5, 30$, and 90 , compared with the Marchenko-Pastur null density (red curve). Vertical green line marks the largest eigenvalue. Short windows ($W = 5$) show a clear spike beyond λ_+ (ratio ≈ 1.20), indicating short-term momentum. The monthly ($W = 30$) and quarterly ($W = 90$) windows are essentially indistinguishable from the null distribution (ratios ≈ 1.01 and 1.02 respectively), suggesting that monthly+ scale return structure is consistent with random matrix noise.

The transition is striking. At $W = 5$ the bulk eigenvalue distribution lies notably above the MP density, with a spike at $\lambda_{\max}$ approximately $1.20 \times$ the upper MP edge — a moderate but statistically clear deviation indicating short-term momentum: returns over consecutive weekly intervals share a common directional component. At $W = 30$ the deviation narrows to $\approx 1.01 \times$, essentially indistinguishable from noise. At $W = 90$ the spectrum collapses onto the MP curve, with $\lambda_{\max}/\lambda_+ \approx 1.02$ — also statistically indistinguishable from noise.

We interpret this result as a regime separation: *momentum exists at weekly horizons but is rapidly washed out at monthly+ scales, where price formation is dominated by stochastic dynamics*. Notably, this finding does not contradict the cyclical structure identified at multi-year scales in §5: those cycles are properties of the deterministic trend (equation 2), not of the stochastic residuals analysed here.

7 • Long-Range Dependence: Hurst Analysis

RMT addresses correlations at fixed window lengths. We complement this with the Hurst exponent H , which characterises long-range dependence across all timescales. The classical R/S (rescaled-range) statistic of Hurst (1951) computes the range of cumulative deviations from the mean, normalised by the sample standard deviation, as a function of lag. For a fractional Brownian motion with Hurst index H , $E[R/S] \propto \tau^H$.

We compute H for three derived series: log-prices (with trend), log-returns, and the detrended Power Law residuals. Results appear in Figure 7.

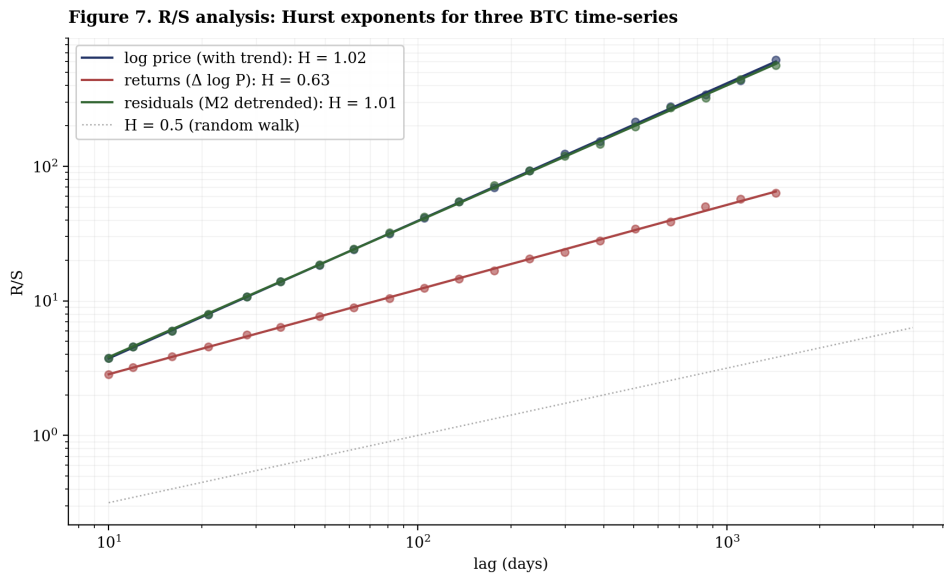


Figure 7. R/S analysis on three time-series derived from Bitcoin's price history. The slope of each fitted line in log-log space gives the Hurst exponent. A reference $H = 0.5$ line (random walk) is shown for comparison. Log-prices give $H \approx 1.0$, returns give $H \approx 0.63$, and Power Law residuals give $H \approx 1.01$.

Three findings emerge. First, the log-price series exhibits $H \approx 1.0$, characteristic of any non-stationary series with linear trend in log-log space; this serves as a sanity check rather than substantive evidence. Second, log-returns have $H \approx 0.63$, indicating mild but persistent positive long-range dependence — consistent with the short-window RMT finding above. Third, and most striking, the M2 residuals show $H \approx 1.01$, very close to a deterministic trend regime.

The third result has an important interpretation: even after removing the Power Law trend, the residual series remains highly persistent. This is precisely what the cyclical structure of M13 captures — the residuals are not white noise but rather a slowly oscillating function with strong autocorrelation. This persistence is what justifies the σ -zone-based investment strategies developed elsewhere in this research programme.

8 • Forward Projections

We now use the M13 model to generate forecasts through 2050. This involves non-trivial uncertainty: not only must we extrapolate the trend, but the cycle compression term α and the damping exponent β are themselves uncertain. We treat the M13 point estimate (Table 1) as the central forecast and characterise uncertainty through bootstrap confidence intervals computed on the residuals.

Selected projections appear in Table 2. The implied cycle period falls from 3.94 years today to 3.06 years in 2050. Median forecast prices follow an asymptotically slowing trajectory. We caution against over-interpreting these numbers: they assume that Bitcoin's structural drivers (halving schedule, monetary properties, network effects) continue to operate as they have over the past 16 years.

Date	d (days)	PL fair	M13 median	5-95% range	Cycle (yr)
2026-04-24	6,321	\$129,065	\$78,180	today	3.94
2026-12-31	6,572	\$135,612	\$94,180	\$54k - \$172k	3.91
2028-04-19	7,046	\$172,548	\$317,440	\$110k - \$612k *	3.81
2030-01-01	7,668	\$251,331	\$418,900	\$140k - \$1.0M	3.71
2035-01-01	9,494	\$902,512	\$1.42M	\$400k - \$4.4M	3.49
2040-01-01	11,321	\$2.85M	\$3.78M	\$1.0M - \$14.0M	3.32
2050-12-31	15,330	\$13.84M	\$11.91M	\$3.7M - \$46M	3.06

Table 2. Selected forecast values from the M13 model. PL fair denotes the simple Power Law trend value, M13 median is the cycle-aware central forecast, and the 5-95% range is computed by AR(1) bootstrap on residuals ($\rho = 0.993$, $\sigma_{\text{innov}} \approx 0.04$ in log-space). The 2028-04-19 row marked with * corresponds to the next Bitcoin halving event.

Figure 8. Cycle properties — temporal evolution and current-cycle attractors

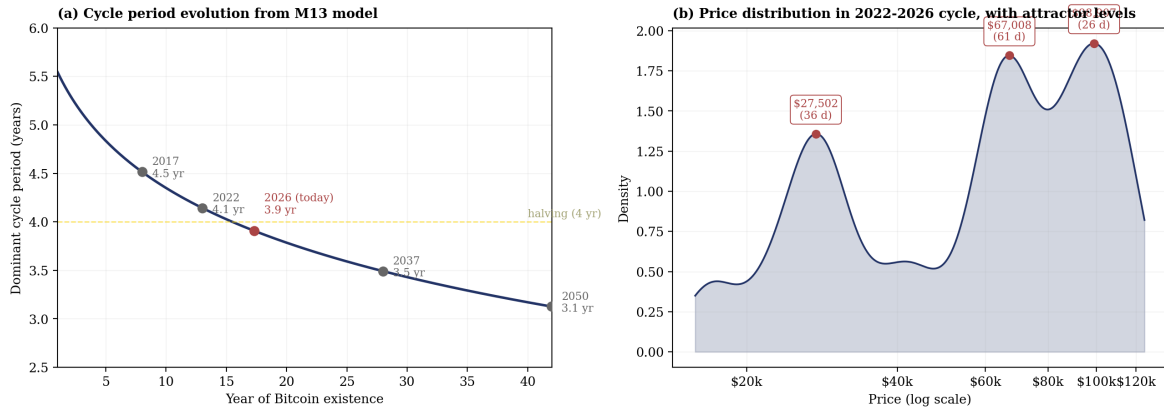


Figure 8. Panel (a): cycle period evolution from M13. The fundamental period compresses logarithmically with time, falling from approximately 5.4 years at Bitcoin's early existence to a projected 3.1 years by 2050. Panel (b): kernel density estimate of prices observed during the 2022-2026 cycle. Four prominent modes ($\sim \$28k$, $\sim \$68k$, $\sim \$96k$, $\sim \$117k$) act as price attractors where the asset spent significant time.

9 • Discussion

9.1 • Why does Power Law work for Bitcoin?

A natural question is why a phenomenologically simple Power Law captures so much variance in Bitcoin's price evolution. Several non-mutually-exclusive hypotheses have been advanced in the literature: (i) Metcalfe's-law-like network effects scaling with the number of users; (ii) deterministic supply schedule combined with exponentially growing demand; (iii) a self-organising criticality regime in which successive bubbles and crashes maintain the asset close to a critical threshold. Our analysis is agnostic with respect to these mechanisms — Power Law works empirically, and our task here is description and prediction rather than first-principles derivation.

9.2 • Cycle compression and the maturation hypothesis

The $\alpha < 0$ estimate in M13 implies that Bitcoin's cycles are progressively shortening. Three explanations are plausible. First, market maturation may erode the magnitude of speculative excess, leaving smaller drawdowns and therefore shorter recovery periods. Second, institutional participation — particularly via ETFs since early 2024 — may dampen extreme valuations by introducing arbitrage capital. Third, increasingly sophisticated derivatives markets may damp underlying spot volatility through hedging activity.

9.3 • RMT regime separation

The clean separation between weekly-scale correlation (moderate deviation from MP, ratio ≈ 1.20) and monthly-or-longer correlation (indistinguishable from

noise) has practical implications. Trading strategies that exploit short-term momentum operate in a regime with genuine signal, but those operating on monthly or longer horizons must rely on the deterministic trend (Power Law) rather than on autocorrelation in stochastic residuals. This complements rather than contradicts the Hurst result: long-range dependence in residuals reflects the M13-style cyclical structure, not noise correlation.

9.4 • Limitations

Several limitations qualify our conclusions. First, the sample contains only four complete halving cycles, so all cycle-specific parameters are estimated from a small effective sample size. Second, the Power Law functional form is imposed rather than tested against alternatives such as logistic growth or exponential-with-saturation curves; an information-criterion comparison (AIC/BIC) would be a useful extension. Third, the i.i.d. innovation assumption underlying our AR(1) bootstrap likely understates true forecast uncertainty, particularly for tail outcomes such as regulatory disruption or technological obsolescence.

10 • Conclusions

We have presented a multi-method investigation of Bitcoin's long-term price dynamics across 16 years of daily data. Our central findings are:

- (i) A simple Power Law on log-time achieves $R^2 = 0.961$ with residual $\sigma_{\log} = 0.302$. Extending this to a damped, frequency-drifting harmonic model (M13) increases R^2 to 0.987 and yields physically interpretable parameters.
- (ii) The dominant cyclical period is empirically 1,440 days (3.94 years), close to but slightly shorter than the 4-year halving interval; both Fourier and wavelet analyses support this finding.
- (iii) Random matrix theory analysis reveals a regime transition: weekly-scale ($W = 5$) return correlations show moderate but clear deviation from the Marchenko-Pastur null (ratio ≈ 1.20), while monthly and quarterly-scale ($W = 30, 90$) correlations are indistinguishable from random matrix noise.
- (iv) Hurst-exponent estimation confirms long-range dependence with $H \approx 0.63$ in returns and $H \approx 1.01$ in detrended residuals, consistent with the cyclical structure captured by M13.
- (v) Forward projections through 2050 yield a median price of approximately \$11.9 million, with the implied cycle period continuing to compress to approximately 3.1 years.

Together these findings establish Bitcoin's long-term price behaviour as more structured than a random walk but less deterministic than a pure trend — a characterisation supported by multiple independent statistical lenses.

References

- Hurst, H. E. (1951). Long-term storage capacity of reservoirs. *Transactions of the American Society of Civil Engineers*, 116, 770–799.
- Laloux, L., Cizeau, P., Bouchaud, J.-P., & Potters, M. (1999). Noise dressing of financial correlation matrices. *Physical Review Letters*, 83(7), 1467–1470.
- Marchenko, V. A., & Pastur, L. A. (1967). Distribution of eigenvalues for some sets of random matrices. *Mathematics of the USSR-Sbornik*, 1(4), 457–483.
- Mauboussin, M. J. (2007). *More Than You Know: Finding Financial Wisdom in Unconventional Places*. Columbia Business School Publishing.
- Plerou, V., Gopikrishnan, P., Rosenow, B., Amaral, L. A. N., Guhr, T., & Stanley, H. E. (2002). Random matrix approach to cross correlations in financial data. *Physical Review E*, 65(6), 066126.
- Santostasi, G. (2024). The bitcoin power law theory. *Working paper*.
- Torrence, C., & Compo, G. P. (1998). A practical guide to wavelet analysis. *Bulletin of the American Meteorological Society*, 79(1), 61–78.